

30 DAYS TO AI ENGINEER

Beginner-Friendly · Project-Driven · Outcome-Focused

30

DAYS

4

WEEKS

4

PROJECTS

20+

RESOURCES

WEEKLY MILESTONES

WEEK 01 · DAYS 1-7

PYTHON & ML FOUNDATIONS

Foundation

- Set up Python environment (conda/venv), Jupyter, VS Code
- Numpy & Pandas** — array ops, DataFrames, data wrangling
- Matplotlib / Seaborn — visualise datasets end-to-end
- Scikit-learn basics** — linear regression, decision trees, evaluation metrics
- Milestone: Train your first ML model on the Titanic dataset

WEEK 02 · DAYS 8-14

DEEP LEARNING & NEURAL NETS

Core AI

- PyTorch fundamentals** — tensors, autograd, training loop
- Build feedforward networks from scratch, understand backprop
- CNNs — image classification on CIFAR-10 / MNIST
- Loss functions, optimisers, batch training, GPU setup
- Milestone: Image classifier reaching 85%+ validation accuracy

WEEK 03 · DAYS 15-22

LLMS, TRANSFORMERS & PROMPTING

LLM Era

- Transformer architecture** — attention, tokens, embeddings
- HuggingFace Transformers & Datasets library
- LangChain + OpenAI / Claude APIs** — chains, agents, tools
- RAG (Retrieval Augmented Generation) — FAISS, ChromaDB
- Milestone: RAG-powered Q&A bot over custom documents

WEEK 04 · DAYS 23-30

DEPLOYMENT & PRODUCTION AI

Ship It

- FastAPI** — wrap model/LLM as REST endpoint
- Streamlit / Gradio — interactive AI demo UIs
- Docker basics — containerise your AI app
- Deploy to HuggingFace Spaces or Render (free tier)
- Milestone: Live public AI app with shareable URL

DAILY SCHEDULE

DAY	FOCUS	TASKS	TIME
Day 1	Environment Setup	Install Python, conda, Jupyter, VS Code extensions setup	2h
Day 2	Numpy	Arrays, broadcasting, matrix ops — complete 20 exercises resource	3h
Day 3	Pandas	Load CSV, groupby, merge, handle nulls on real dataset core	3h
Day 4	Visualisation	EDA on Titanic dataset — 5 chart types, seaborn heatmaps	2.5h
Day 5	Scikit-learn I	Train/test split, regression, classification pipeline core	3h
Day 6	Scikit-learn II	Cross-validation, hyperparameter tuning, feature engineering	3h

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Day 7	Project 1 	Build end-to-end Titanic survival predictor + notebook report project	4h
Day 8	PyTorch Basics	Tensors, autograd, simple operations; compare with NumPy core	3h
Day 9	Neural Network I	Build MLP from scratch, understand forward pass + backprop	3h
Day 10	Training Loop	Loss, optimiser, batch loader, evaluation loop on MNIST core	3h
Day 11	CNNs	Conv layers, pooling, LeNet-5 from scratch on CIFAR-10	3.5h
Day 12	Transfer Learning	Fine-tune ResNet18 on custom 5-class image dataset resource	3h
Day 13	Regularisation	Dropout, BatchNorm, augmentation — improve accuracy 5%+	3h
Day 14	Project 2 	Deploy image classifier as standalone Python script project	4h
Day 15	Transformer Theory	Read "Attention is All You Need" summary; visualise attention heads resource	2h
Day 16	HuggingFace	Load BERT, run sentiment analysis, explore model hub core	3h
Day 17	OpenAI / Claude API	Make API calls, system prompts, temperature, JSON mode	3h
Day 18	Prompt Engineering	Chain-of-thought, few-shot, role prompts — build a prompt library	2.5h
Day 19	LangChain	Chains, memory, tools — build a multi-step agent core	3h
Day 20	Embeddings + FAISS	Text embeddings, cosine similarity, vector search basics	3h
Day 21	RAG Pipeline	Chunk PDF → embed → store → retrieve → LLM answer core	3.5h
Day 22	Project 3 	RAG chatbot over a chosen PDF / knowledge base project	4h
Day 23	FastAPI	Build REST API endpoint for your RAG bot; test with Postman core	3h
Day 24	Streamlit UI	Add chat UI to the RAG bot; upload-PDF feature	3h
Day 25	Docker	Dockerise the full app; write Dockerfile + docker-compose core	3h
Day 26	Cloud Deploy I	Push to HuggingFace Spaces or Render; environment secrets	3h
Day 27	Cloud Deploy II	Add GitHub Actions CI/CD — auto-deploy on push	2.5h
Day 28	Polish & Debug	Fix bugs, add loading states, error handling, README resource	3h
Day 29	Portfolio	Write LinkedIn post + GitHub README for each project	2.5h
Day 30	Project 4 	Launch final capstone — share live URL publicly project	4h

+ RESOURCES

COURSE

fast.ai — Practical Deep Learning

Top-down, code-first. Free. Ideal for beginners who learn by doing.

COURSE

DeepLearning.AI Specialisation

Andrew Ng's classic. Covers ML fundamentals through deep learning rigorously.

BOOK

Hands-On ML — Aurélien Géron

The definitive practical ML book. Covers Scikit-learn, TensorFlow/Keras deeply.

YOUTUBE

Andrej Karpathy — Neural Nets: Zero to Hero

Builds GPT from scratch. Best deep-dive into transformers for beginners.

COURSE

HuggingFace NLP Course

Free. Covers Transformers, tokenisers, fine-tuning, and the HF ecosystem.

DOCS

LangChain Documentation

Official docs for chains, agents, RAG pipelines, and integrations.

COURSE

Full Stack LLM Bootcamp

Berkeley's free course covering prompt engineering to deployment. Very practical.

PRACTICE

Kaggle Learn + Competitions

Free micro-courses + real competitions. Build portfolio credibility fast.

TOOL

Weights & Biases (W&B)

Experiment tracking, model versioning — used at top AI labs. Free tier.

PLATFORM

Google Colab (Free GPU)

Run notebooks with T4 GPU free. Essential if you don't have local GPU.

PAPER

Attention Is All You Need (2017)

The original Transformer paper. Read the summary on Papers With Code first.

YOUTUBE

Sentdex & AssemblyAI Channel

Practical ML/AI tutorials. Great for PyTorch, APIs, and deployment guides.

WEEK 1 · DAY 7

01

TITANIC SURVIVAL PREDICTOR

End-to-end ML pipeline: data cleaning, feature engineering, model training & evaluation using classic Titanic dataset. Export a reusable inference function.

pandas

scikit-learn

matplotlib

Jupyter

WEEK 2 · DAY 14

02

IMAGE CLASSIFIER (FINE-TUNED)

Fine-tune ResNet18 on a custom 5-class image dataset (e.g. food, animals). Achieve 85%+ accuracy and wrap as a Python inference script.

PyTorch

torchvision

transfer-learning

CNN

WEEK 3 · DAY 22

03

RAG DOCUMENT CHATBOT

Upload any PDF → chunk & embed → store in FAISS → ask questions. Powered by LangChain + OpenAI/Claude API. Accurate answers grounded in source docs.

LangChain

FAISS

OpenAI API

embeddings

WEEK 4 · DAY 30

04

DEPLOYED AI WEB APP (CAPSTONE)

Full-stack AI product: Streamlit/FastAPI frontend, Docker container, deployed live on HuggingFace Spaces or Render. Public URL + GitHub repo + README.

FastAPI

Streamlit

Docker

CI/CD

HuggingFace

✦ FINAL OUTCOME

YOU ARE NOW AN AI ENGINEER

After 30 days you will have built 4 real AI projects, command the modern AI/LLM stack, and have a public portfolio — ready to apply for junior AI engineer or ML engineer roles.



Deep Learning

PyTorch, CNNs, transfer learning, training loops



LLM Engineering

RAG, agents, prompt engineering, vector DBs



Production Skills

FastAPI, Docker, CI/CD, cloud deployment



Portfolio Ready

4 public projects, GitHub, live demo URLs



ML Fundamentals

Data wrangling, evaluation, scikit-learn pipelines



Job-Ready

Junior AI/ML Engineer · AI Developer · LLM Engineer